

Consider the following laminates:

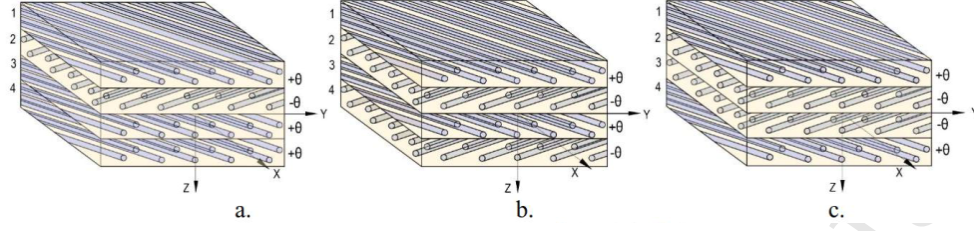


Figure 1: Layer stacking schematic of the laminates

- a. $[+\theta/(\theta)]$ general asymmetric laminate;
- b. $[(\pm\theta)_2]$ antisymmetric angle-ply laminate;
- c. $[\pm\theta]_s$ symmetric angle-ply laminate.

We know that:

$\eta_{i,ij}$ represents the *in-plane coefficient of mutual influence of the first kind*, which describes the stretching in the i direction induced by the shear stress in the ij plane.

$\eta_{ij,i}$ represents the *in-plane coefficient of mutual influence of the second kind*, which describes the shearing in the ij plane induced by the normal stress in the i direction.

$$\eta_{x,xy} = \frac{\varepsilon_x^0}{\gamma_{xy}^0}, \quad \eta_{y,xy} = \frac{\varepsilon_y^0}{\gamma_{xy}^0}, \quad \text{when } \tau_{xy} \neq 0, \sigma_x = 0, \sigma_y = 0. \quad (1a)$$

$$\eta_{xy,x} = \frac{\gamma_{xy}^0}{\varepsilon_x^0}, \quad \text{when } \sigma_x \neq 0, \sigma_y = 0, \tau_{xy} = 0. \quad (1b)$$

$$\eta_{xy,y} = \frac{\gamma_{xy}^0}{\varepsilon_y^0}, \quad \text{when } \sigma_y \neq 0, \sigma_x = 0, \tau_{xy} = 0. \quad (1c)$$

Q1: For general composite laminates

$$\begin{Bmatrix} N_x \\ N_y \\ N_{xy} \end{Bmatrix} = \begin{bmatrix} A_{11} & A_{12} & A_{16} \\ A_{12} & A_{22} & A_{26} \\ A_{16} & A_{26} & A_{66} \end{bmatrix} \begin{Bmatrix} \varepsilon_x^0 \\ \varepsilon_y^0 \\ \gamma_{xy}^0 \end{Bmatrix} + \begin{bmatrix} B_{11} & B_{12} & B_{16} \\ B_{12} & B_{22} & B_{26} \\ B_{16} & B_{26} & B_{66} \end{bmatrix} \begin{Bmatrix} k_x \\ k_y \\ k_{xy} \end{Bmatrix} \quad (2a)$$

$$\begin{Bmatrix} M_x \\ M_y \\ M_{xy} \end{Bmatrix} = \begin{bmatrix} B_{11} & B_{12} & B_{16} \\ B_{12} & B_{22} & B_{26} \\ B_{16} & B_{26} & B_{66} \end{bmatrix} \begin{Bmatrix} \varepsilon_x^0 \\ \varepsilon_y^0 \\ \gamma_{xy}^0 \end{Bmatrix} + \begin{bmatrix} D_{11} & D_{12} & D_{16} \\ D_{12} & D_{22} & D_{26} \\ D_{16} & D_{26} & D_{66} \end{bmatrix} \begin{Bmatrix} k_x \\ k_y \\ k_{xy} \end{Bmatrix} \quad (2b)$$

show that

$$\eta_{x,xy} = \frac{\begin{vmatrix} A_{12} & A_{16} & B_{11} & B_{12} & B_{16} \\ A_{22} & A_{26} & B_{12} & B_{22} & B_{26} \\ B_{12} & B_{16} & D_{11} & D_{12} & D_{16} \\ B_{22} & B_{26} & D_{12} & D_{22} & D_{26} \\ B_{26} & B_{66} & D_{16} & D_{26} & D_{66} \end{vmatrix}}{\begin{vmatrix} A_{11} & A_{12} & B_{11} & B_{12} & B_{16} \\ A_{12} & A_{22} & B_{12} & B_{22} & B_{26} \\ B_{11} & B_{12} & D_{11} & D_{12} & D_{16} \\ B_{12} & B_{22} & D_{12} & D_{22} & D_{26} \\ B_{16} & B_{26} & D_{16} & D_{26} & D_{66} \end{vmatrix}} \quad \eta_{xy,x} = \frac{\begin{vmatrix} A_{12} & A_{22} & B_{12} & B_{22} & B_{26} \\ A_{16} & A_{26} & B_{16} & B_{26} & B_{66} \\ B_{11} & B_{12} & D_{11} & D_{12} & D_{16} \\ B_{12} & B_{22} & D_{12} & D_{22} & D_{26} \\ B_{16} & B_{26} & D_{16} & D_{26} & D_{66} \end{vmatrix}}{\begin{vmatrix} A_{22} & A_{26} & B_{12} & B_{22} & B_{26} \\ A_{26} & A_{66} & B_{16} & B_{26} & B_{66} \\ B_{12} & B_{16} & D_{11} & D_{12} & D_{16} \\ B_{22} & B_{26} & D_{12} & D_{22} & D_{26} \\ B_{26} & B_{66} & D_{16} & D_{26} & D_{66} \end{vmatrix}}$$

Solution:

From the definitions of force and moment resultants

$$N_x = \int_{-H/2}^{H/2} \sigma_x dz, \quad N_y = \int_{-H/2}^{H/2} \sigma_y dz, \quad N_{xy} = \int_{-H/2}^{H/2} \tau_{xy} dz.$$

$$M_x = \int_{-H/2}^{H/2} \sigma_x z dz, \quad M_y = \int_{-H/2}^{H/2} \sigma_y z dz, \quad M_{xy} = \int_{-H/2}^{H/2} \sigma_z z dz.$$

by considering equ. (1a) for $\eta_{x,xy}$,

$$\eta_{x,xy} = \frac{1}{E_x} \left(\frac{N_x}{H} + \frac{M_x}{H^2} \right) + \frac{1}{E_y} \left(\frac{N_y}{H} + \frac{M_y}{H^2} \right) + \frac{1}{G_{xy}} \left(\frac{N_{xy}}{H} + \frac{M_{xy}}{H^2} \right)$$

and by considering equ. (1b) for $\eta_{xy,x}$,

$$\eta_{xy,x} = \frac{1}{E_x} \left(\frac{N_x}{H} + \frac{M_x}{H^2} \right) + \frac{1}{E_y} \left(\frac{N_y}{H} + \frac{M_y}{H^2} \right) + \frac{1}{G_{xy}} \left(\frac{N_{xy}}{H} + \frac{M_{xy}}{H^2} \right)$$

From (2a) and (2b) we write the ABD matrix as

$$\begin{Bmatrix} N_x \\ N_y \\ N_{xy} \\ M_x \\ M_y \\ M_{xy} \end{Bmatrix} = \begin{bmatrix} A_{11} & A_{12} & A_{16} & B_{11} & B_{12} & B_{16} \\ A_{12} & A_{22} & A_{26} & B_{12} & B_{22} & B_{26} \\ A_{16} & A_{26} & A_{66} & B_{16} & B_{26} & B_{66} \\ B_{11} & B_{12} & B_{16} & D_{11} & D_{12} & D_{16} \\ B_{12} & B_{22} & B_{26} & D_{12} & D_{22} & D_{26} \\ B_{16} & B_{26} & B_{66} & D_{16} & D_{26} & D_{66} \end{bmatrix} \begin{Bmatrix} \epsilon_x^0 \\ \epsilon_y^0 \\ \gamma_{xy}^0 \\ \kappa_x^0 \\ \kappa_y^0 \\ \kappa_{xy}^0 \end{Bmatrix}$$

Case 1: $\eta_{x,xy}$

Consider the following loading conditions:

We obtain

$$\eta_{x,xy} = \frac{\varepsilon_x^0}{\gamma_{xy}^0} = \frac{\begin{vmatrix} A_{12} & A_{16} & B_{11} & B_{12} & B_{16} \\ A_{22} & A_{26} & B_{12} & B_{22} & B_{26} \\ B_{12} & B_{16} & D_{11} & D_{12} & D_{16} \\ B_{22} & B_{26} & D_{12} & D_{22} & D_{26} \\ B_{26} & B_{66} & D_{16} & D_{26} & D_{66} \end{vmatrix}}{\begin{vmatrix} A_{11} & A_{12} & B_{11} & B_{12} & B_{16} \\ A_{12} & A_{22} & B_{12} & B_{22} & B_{26} \\ B_{11} & B_{12} & D_{11} & D_{12} & D_{16} \\ B_{12} & B_{22} & D_{12} & D_{22} & D_{26} \\ B_{16} & B_{26} & D_{16} & D_{26} & D_{66} \end{vmatrix}} \quad (3)$$

Case 2: $\eta_{xy,x}$

Consider the following loading conditions:

We obtain

$$\eta_{xy,x} = \begin{bmatrix} A_{12} & A_{22} & B_{12} & B_{22} & B_{26} \\ A_{16} & A_{26} & B_{16} & B_{26} & B_{66} \\ B_{11} & B_{12} & D_{11} & D_{12} & D_{16} \\ B_{12} & B_{22} & D_{12} & D_{22} & D_{26} \\ B_{16} & B_{26} & D_{16} & D_{26} & D_{66} \end{bmatrix} \begin{bmatrix} A_{22} & A_{26} & B_{12} & B_{22} & B_{26} \\ A_{26} & A_{66} & B_{16} & B_{26} & B_{66} \\ B_{12} & B_{16} & D_{11} & D_{12} & D_{16} \\ B_{22} & B_{26} & D_{12} & D_{22} & D_{26} \\ B_{26} & B_{66} & D_{16} & D_{26} & D_{66} \end{bmatrix} \quad (4)$$

Q2: For symmetric angle-ply laminates:

$$\begin{pmatrix} \varepsilon_x^0 \\ \varepsilon_y^0 \\ \gamma_{xy}^0 \end{pmatrix} = \begin{bmatrix} A'_{11} & A'_{12} & A'_{16} \\ A'_{12} & A'_{22} & A'_{26} \\ A'_{16} & A'_{26} & A'_{66} \end{bmatrix} \begin{pmatrix} N_x \\ N_y \\ N_{xy} \end{pmatrix} \text{ or } \begin{pmatrix} N_x \\ N_y \\ N_{xy} \end{pmatrix} = \begin{bmatrix} A_{11} & A_{12} & A_{16} \\ A_{12} & A_{22} & A_{26} \\ A_{16} & A_{26} & A_{66} \end{bmatrix} \begin{pmatrix} \varepsilon_x^0 \\ \varepsilon_y^0 \\ \gamma_{xy}^0 \end{pmatrix} \quad (5)$$

show that:

$$\eta_{xy,x} = \frac{A'_{16}}{A'_{11}}, \quad \eta_{x,xy} = \frac{A'_{16}}{A'_{66}}$$

Plot these variations with respect to the fiber angle θ for S-Glass/Epoxy with fiber volume fraction $V_f = 0.7$.

Table 1: Constituents of composite plies and their properties

Constituent	Longitudinal Young's Modulus (GPa)	Poisson's Ratio
S-glass fibres	85.5	0.22
E-glass fibres	72.4	0.22
Epoxy resin matrix	4.1	0.40

Solution: In symmetric laminates, the B matrix becomes zero; therefore, to calculate the coefficients, only the A matrix is required.

Case 1: $\eta_{x,xy}$

Consider the following loading conditions:

$$\eta_{x,xy} = \frac{A'_{16}}{A'_{66}} \quad (6)$$

Case 1: $\eta_{xy,x}$

Consider the following loading conditions:

$$\eta_{xy,x} = \frac{A'_{16}}{A'_{11}} \quad (7)$$

Next, we need to determine the material properties of the S-Glass/Epoxy composite. To compute A_{ij} , we use:

$$A_{ij} = \sum_{k=1}^N \bar{Q}_{ij_k} (z_k - z_{k-1}) \quad (8)$$

For the transformed stiffness coefficients, where $m = \cos \theta$ and $n = \sin \theta$, we have

$$\bar{Q}_{11} = Q_{11}m^4 + 2(Q_{12} + 2Q_{66})m^2n^2 + Q_{22}n^4 \quad (9a)$$

$$\bar{Q}_{12} = (Q_{11} + Q_{22} - 4Q_{66})n^2m^2 + Q_{12}(n^4 + m^4) \quad (9b)$$

$$\bar{Q}_{16} = (Q_{11} - Q_{12} - 2Q_{66})nm^3 + (Q_{12} - Q_{22} + 2Q_{66})n^3m \quad (9c)$$

$$\bar{Q}_{22} = Q_{11}n^4 + 2(Q_{12} + 2Q_{66})n^2m^2 + Q_{22}m^4 \quad (9d)$$

$$\bar{Q}_{26} = (Q_{11} - Q_{12} - 2Q_{66})n^3m + (Q_{12} - Q_{22} + 2Q_{66})nm^3 \quad (9e)$$

$$\bar{Q}_{66} = (Q_{11} + Q_{22} - 2Q_{12} - 2Q_{66})n^2m^2 + Q_{66}(n^4 + m^4) \quad (9f)$$

$$(9g)$$

and for Q_{ij} :

$$Q_{11} = \frac{E_1}{1 - \nu_{12}\nu_{21}} \quad (10a)$$

$$Q_{12} = \frac{\nu_{12}E_2}{1 - \nu_{12}\nu_{21}} = \frac{\nu_{21}E_1}{1 - \nu_{12}\nu_{21}} \quad (10b)$$

$$Q_{22} = \frac{E_2}{1 - \nu_{12}\nu_{21}} \quad (10c)$$

$$Q_{66} = G_{12} \quad (10d)$$

Therefore, the engineering constants E_1 , E_2 , G_{12} , ν_{12} , and ν_{21} are determined using the rule of mixtures.

$$E_1 = E_1^f V^f + E^m V^m \quad (11a)$$

$$\nu_{12} = \nu^f V^f + \nu^m V^m \quad (11b)$$

$$\frac{1}{E_2} = \frac{\eta^f V^f}{E^f} + \frac{\eta^m V^m}{E^m} \quad (11c)$$

$$\eta^f = \frac{E^f V^f + [(1 - (\nu^f)^2)E^m + \nu^m \nu^f E^f] (1 - V^f)}{E^f V^f + E^m V^m} \quad (11d)$$

$$\eta^m = \frac{[(1 - (\nu^m)^2)E^f - (1 - \nu^m \nu^f)E^m] V^f + E^m V^m}{E^f V^f + E^m V^m} \quad (11e)$$

$$\frac{1}{G_{12}} = \frac{V^f}{G^f} + \frac{\eta' V^m}{G^m}, \quad \eta' = 0.6 \quad (11f)$$

$$G = \frac{E}{2(1 + \nu)} \quad (11g)$$

$$\nu_{21} = \nu_{12} \frac{E_2}{E_1} \quad (11h)$$

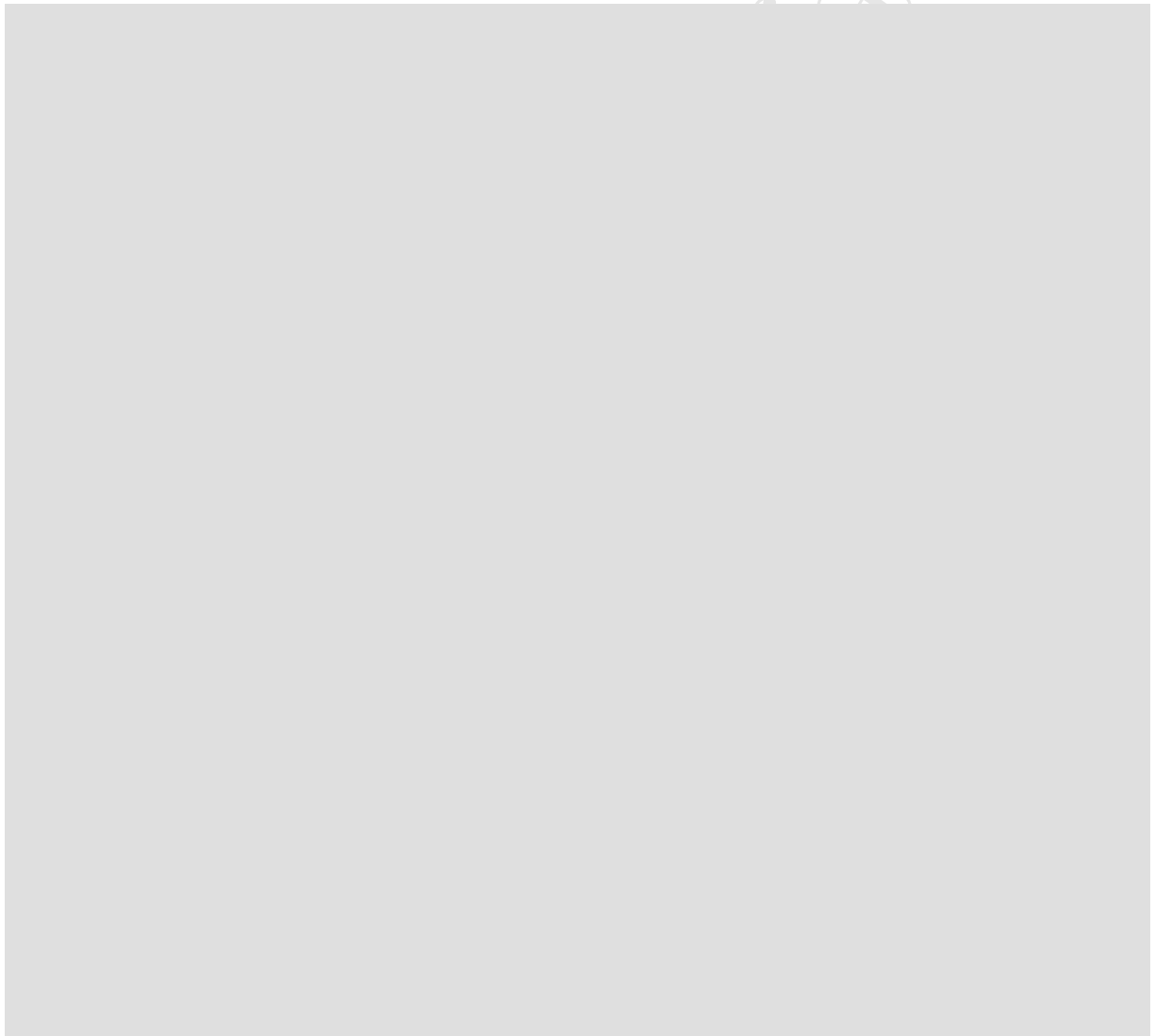
From Table 1, the material properties are

$$E^f = 85.5, \text{ GPa}, \quad \nu^f = 0.22, \quad E^m = 4.1, \text{ GPa}, \quad \nu^m = 0.40, \quad V^f = 0.7, \quad V^m = 0.3.$$

The shear modulus of the fiber and matrix are:

$$G^f = \frac{E^f}{2(1 + \nu^f)} = \frac{85.5}{2(1 + 0.22)} = 35.04 \text{ GPa}$$

$$G^m = \frac{E^m}{2(1 + \nu^m)} = \frac{4.1}{2(1 + 0.4)} = 1.464 \text{ GPa}$$



$$E_1 = 61.08 \text{ GPa}, \quad E_2 = 14.96 \text{ GPa}, \quad \nu_{12} = 0.274, \quad \nu_{21} = 0.0671, \quad G_{12} = 6.156 \text{ GPa}.$$

$$Q_{11} = 62.24 \text{ GPa}, \quad Q_{12} = 4.178 \text{ GPa}, \quad Q_{22} = 15.24 \text{ GPa}, \quad Q_{66} = 6.156 \text{ GPa}.$$

The A9 matrix for a single-layer laminate is computed as

$$A_{ij} = \sum_{k=1}^N \bar{Q}_{ij} (z_k - z_{k-1}) = t \bar{Q}_{ij} \quad i, j = 1, 2, 6$$

$$\begin{bmatrix} A_{11} & A_{12} & A_{16} \\ A_{12} & A_{22} & A_{26} \\ A_{16} & A_{26} & A_{66} \end{bmatrix} = t \begin{bmatrix} \bar{Q}_{11} & \bar{Q}_{12} & \bar{Q}_{16} \\ \bar{Q}_{12} & \bar{Q}_{22} & \bar{Q}_{26} \\ \bar{Q}_{16} & \bar{Q}_{26} & \bar{Q}_{66} \end{bmatrix}$$

$$\bar{Q}_{11}(\theta) = 62.24 m^4 + 32.98 m^2 n^2 + 15.24 n^4$$

$$\bar{Q}_{12}(\theta) = 52.856 n^2 m^2 + 4.178(n^4 + m^4)$$

$$\bar{Q}_{16}(\theta) = (45.75)nm^3 + (1.25)n^3m$$

$$\bar{Q}_{22}(\theta) = 62.24 n^4 + 32.98 n^2 m^2 + 15.24 m^4$$

$$\bar{Q}_{26}(\theta) = 45.75 n^3 m + 1.25 n m^3$$

$$\bar{Q}_{66}(\theta) = 56.812 n^2 m^2 + 6.156(n^4 + m^4)$$

$$A' = \frac{1}{t} \begin{bmatrix} 62.24 m^4 + 32.98 m^2 n^2 + 15.24 n^4 & 52.856 n^2 m^2 + 4.178(n^4 + m^4) & (45.75) n m^3 + (1.25) n^3 m \\ 52.856 n^2 m^2 + 4.178(n^4 + m^4) & 62.24 n^4 + 32.98 n^2 m^2 + 15.24 m^4 & 45.75 n^3 m + 1.25 n m^3 \\ (45.75) n m^3 + (1.25) n^3 m & 45.75 n^3 m + 1.25 n m^3 & 56.812 n^2 m^2 + 6.156(n^4 + m^4) \end{bmatrix}^{-1}$$

$$\eta_{x,xy} = \frac{A'_{16}}{A'_{66}}, \quad \eta_{xy,x} = \frac{A'_{16}}{A'_{11}}$$

Due to the large amount of calculations required for matrix inversion and coefficient extraction, Python code was used to perform the numerical computations and generate the plots.

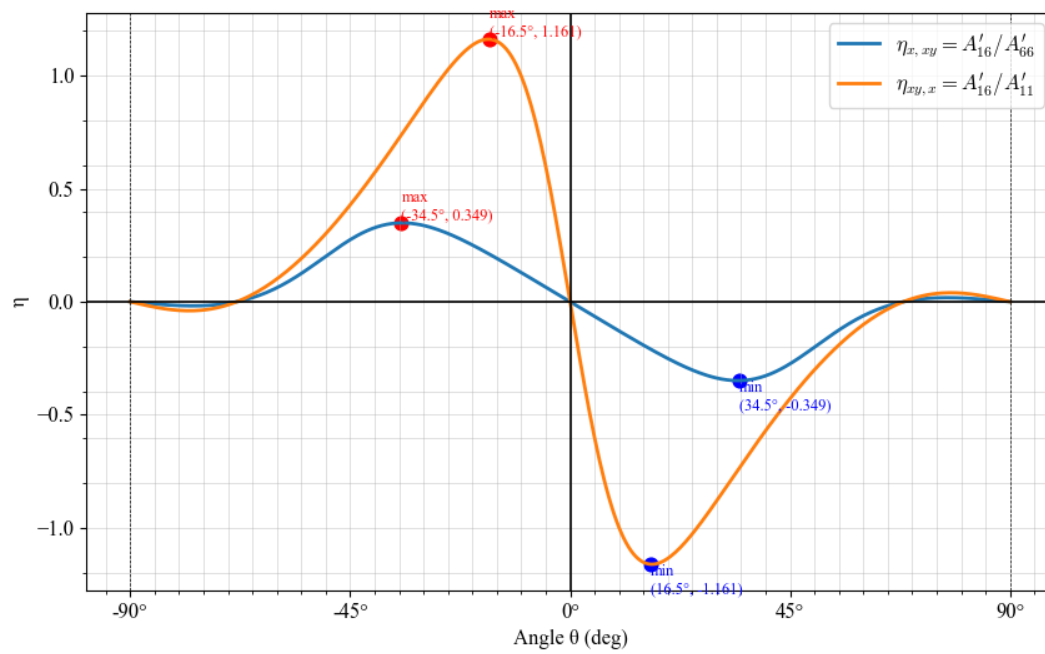


Figure 2: Coupling Coefficients vs. Fiber Angle